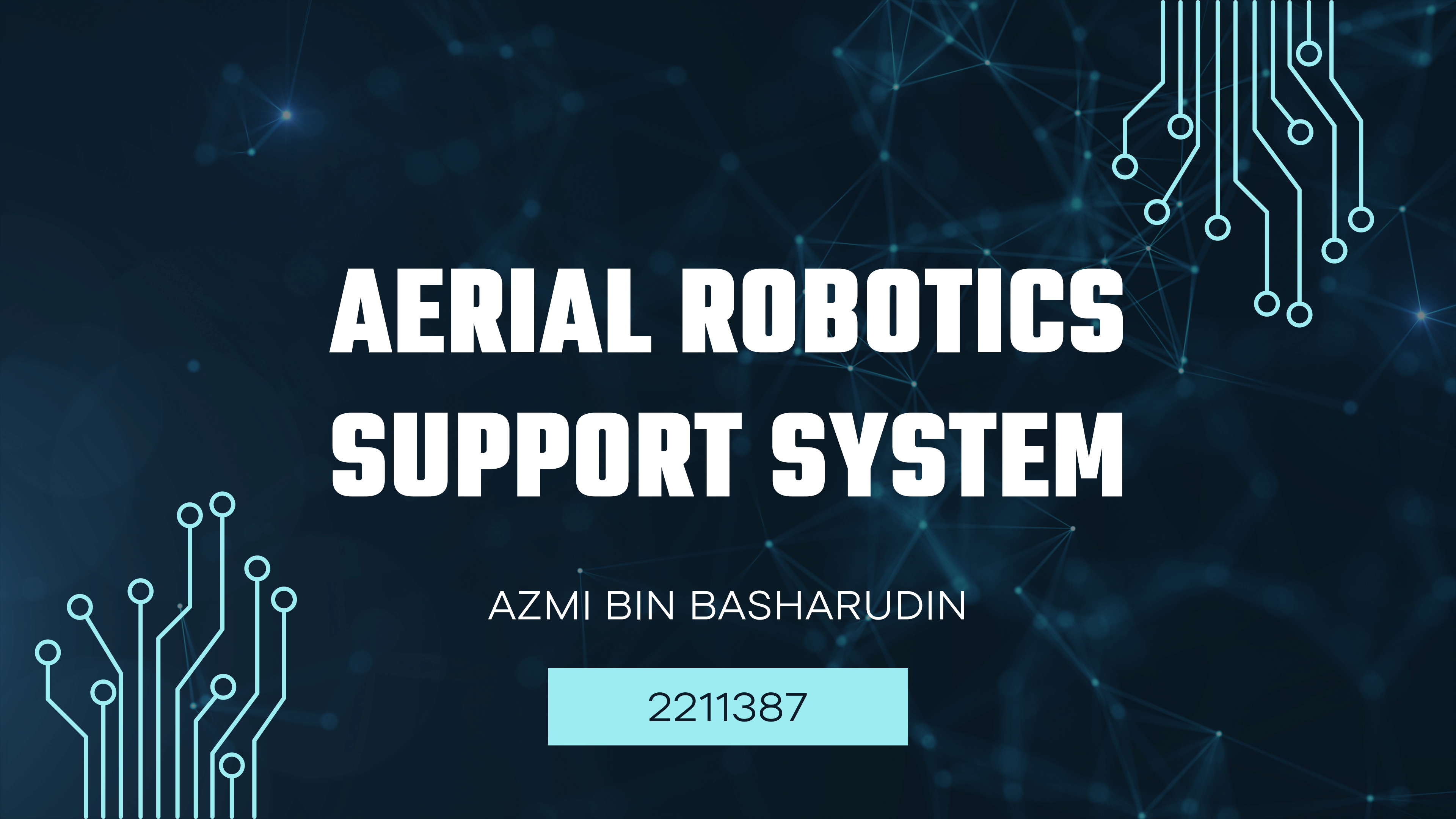




AERIAL ROBOTICS SUPPORT SYSTEM

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2211387

Types of Sensors



- Inertial Measurement Units (IMU): Measures acceleration, angular velocity, orientation.
- GPS Modules: Location tracking; needs clear satellite signal.
- LiDAR (Light Detection and Ranging): 3D environmental mapping.
- Ultrasonic and Infrared Sensors: Obstacle detection at close range.
- Vision Cameras: Object detection, classification using AI.



Alternative Power Systems



- Battery (Li-Ion / Li-Po): Lightweight, rechargeable; limited by runtime.
- Solar Power: Long-term sustainable, but weather-dependent.
- Fuel Cells: Provide higher energy density; suitable for long-duration missions.
- Hybrid Systems: Combine solar + battery or battery + combustion.



Human-Machine Interface (HMI)



- Remote Control: Joystick or gamepad for basic maneuvering.
- Graphical Interfaces (UI): Map overlays, sensor data, manual overrides.
- Voice and Gesture Control: AI-powered, emerging technology.
- Augmented/Virtual Reality (AR/VR): Used in simulations and remote operation environments.



Thank You!

